

دليل تحديث البيانات والتحكم التفاعلي (Live Demo)

مشروع إدارة الامتثال الحوكمي التشريعي — SAMA CSF & SDAIA AI Ethics

الميزات المضافة في التحديث المتكامل:

- القائمة التفاعلية (Interactive Menu): خيارات متعددة لعرض التقرير الكامل، إدراج ضوابط جديدة، وتحديث حالات الامتثال تلقائياً.
- التحديث التفاعلي للمستندات (Live Database Update): تعديل حالة أي ضابط (مثل تغيير حالة SAMA-2.2.1 إلى Compliant) واستعلام النتيجة فوراً.
- معالجة المعاملات (PreparedStatement): حماية كاملة من ثغرات SQL Injection لضمان أعلى معايير الأمان البرمجي.

كود Java التفاعلي المتكامل (`Laab.java`):

```
package com.mycompany.laab;

import java.sql.*;
import java.util.Scanner;

public class Laab {
    private static final String URL = "jdbc:mysql://localhost:3306/compliance_db?zeroDateTimeBehavior=CONVERT_TO_NULL";
    private static final String USER = "root";
    private static final String PASSWORD = "";

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        while (true) {
            System.out.println("\n=====");
            System.out.println("  COMPLIANCE MANAGEMENT SYSTEM - INTERACTIVE SYSTEM  ");
            System.out.println("=====");
            System.out.println("1. View Executive Compliance Dashboard");
            System.out.println("2. Update Control Compliance Status");
            System.out.println("3. Add New Regulatory Control");
            System.out.println("4. Exit");
            System.out.print("Select an option (1-4): ");

            int choice = scanner.nextInt();
            scanner.nextLine();

            switch (choice) {
                case 1 -> displayDashboard();
                case 2 -> updateControlStatus(scanner);
                case 3 -> addNewControl(scanner);
                case 4 -> {
                    System.out.println("Exiting System... Wish you best of luck in the competition!");
                }
            }
        }
    }
}
```

```

        return;
    }
    default -> System.out.println("Invalid option. Try again.");
}
}
}

private static void displayDashboard() {
    String query = "SELECT f.name AS Framework, c.control_code AS Code, c.title AS
Control_Title, "
        + "a.status AS Compliance_Status, a.risk_level AS Risk "
        + "FROM controls c "
        + "JOIN frameworks f ON c.framework_id = f.framework_id "
        + "LEFT JOIN compliance_assessments a ON c.control_id = a.control_id";

    try (Connection conn = DriverManager.getConnection(URL, USER, PASSWORD);
        Statement stmt = conn.createStatement();
        ResultSet rs = stmt.executeQuery(query)) {

        System.out.println("\n=====
=====");

        System.out.println("                                COMPLIANCE MANAGEMENT SYSTEM
DASHBOARD                                ");

        System.out.println("=====
=====");

        System.out.printf("%-18s | %-12s | %-28s | %-18s | %-8s\n", "Framework", "Code",
"Title", "Status", "Risk");

        System.out.println("-----
-----");

        while (rs.next()) {
            System.out.printf("%-18s | %-12s | %-28s | %-18s | %-8s\n",
                rs.getString("Framework"),
                rs.getString("Code"),
                rs.getString("Control_Title"),
                rs.getString("Compliance_Status"),
                rs.getString("Risk"));
        }

        System.out.println("=====
=====");
    } catch (Exception e) {
        System.out.println("Error loading dashboard: " + e.getMessage());
    }
}

private static void updateControlStatus(Scanner scanner) {
    System.out.print("Enter Control Code (e.g. SAMA-2.2.1): ");
    String code = scanner.nextLine();
    System.out.print("Enter New Status (Compliant / Non-Compliant / Under Review /
Partially Compliant): ");
    String status = scanner.nextLine();

    String updateQuery = "UPDATE compliance_assessments a "
        + "JOIN controls c ON a.control_id = c.control_id "
        + "SET a.status = ? WHERE c.control_code = ?";

    try (Connection conn = DriverManager.getConnection(URL, USER, PASSWORD);

```

```

        PreparedStatement pstmt = conn.prepareStatement(updateQuery)) {

    pstmt.setString(1, status);
    pstmt.setString(2, code);

    int rowsAffected = pstmt.executeUpdate();
    if (rowsAffected > 0) {
        System.out.println(">>> SUCCESS: Status updated successfully for " + code);
    } else {
        System.out.println(">>> WARNING: Control Code not found.");
    }
} catch (Exception e) {
    System.out.println("Error updating status: " + e.getMessage());
}
}

private static void addNewControl(Scanner scanner) {
    System.out.print("Enter Framework ID (1 for SAMA CSF, 2 for SDAIA AI Ethics): ");
    int frameworkId = scanner.nextInt();
    scanner.nextLine();
    System.out.print("Enter Control Code (e.g. SAMA-4.1.1): ");
    String code = scanner.nextLine();
    System.out.print("Enter Domain Title: ");
    String domain = scanner.nextLine();
    System.out.print("Enter Control Title: ");
    String title = scanner.nextLine();
    System.out.print("Enter Description: ");
    String desc = scanner.nextLine();

    String insertControl = "INSERT INTO controls (framework_id, control_code, domain,
title, description) VALUES (?, ?, ?, ?, ?)";
    String insertAssessment = "INSERT INTO compliance_assessments (control_id, status,
risk_level) VALUES (?, 'Under Review', 'Medium')";

    try (Connection conn = DriverManager.getConnection(URL, USER, PASSWORD);
        PreparedStatement pstmt1 = conn.prepareStatement(insertControl,
Statement.RETURN_GENERATED_KEYS)) {

        pstmt1.setInt(1, frameworkId);
        pstmt1.setString(2, code);
        pstmt1.setString(3, domain);
        pstmt1.setString(4, title);
        pstmt1.setString(5, desc);
        pstmt1.executeUpdate();

        ResultSet rs = pstmt1.getGeneratedKeys();
        if (rs.next()) {
            int newControlId = rs.getInt(1);
            try (PreparedStatement pstmt2 = conn.prepareStatement(insertAssessment)) {
                pstmt2.setInt(1, newControlId);
                pstmt2.executeUpdate();
            }
        }
        System.out.println(">>> SUCCESS: New control added and initialized!");
    } catch (Exception e) {
        System.out.println("Error adding control: " + e.getMessage());
    }
}
}

```